Week 8: Review of Weeks 1–7

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Session 8.2 Homework

Instructions: Show all work. Circle your final answers.

Topics Covered:

- Week 2: Solving Linear Equations
- Week 3: Inequalities (Absolute Value)
- Week 5: Systems of Linear Equations
- Week 6: Linear Functions & Lines (Parallel/Perpendicular)
- Week 7: Systems Applications & Inequalities

Quick Reference: Key Formulas & Concepts

Absolute Value Inequalities

For $|x| < a$ (where $a > 0$): $-a < x < a$
For $|x| > a$ (where $a > 0$): $x < -a$ OR $x > a$

Parallel & Perpendicular Lines

Parallel lines: Same slope ($m_1 = m_2$)
Perpendicular lines: Negative reciprocal slopes ($m_1 \cdot m_2 = -1$)

Systems of Equations

One solution: Lines intersect (different slopes)
No solutions: Lines are parallel (same slope, different intercepts)
Infinitely many solutions: Same line

Distance Formula (Motion)

$d = rt \quad$ (distance = rate × time)

Mixture Problems

$(\text{concentration}_1)(\text{volume}_1) + (\text{concentration}_2)(\text{volume}_2) = (\text{concentration}_{\text{final}})(\text{volume}_{\text{final}})$

Part A: Equations & Special Cases

Problem 1

Solve: $3(x-4) = 3(x+7)$

Problem 2

Solve: $2(x+5) - 3 = 2x + 7$

Problem 3

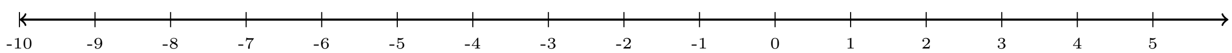
Solve: $\dfrac{x}{4} + 5 = \dfrac{x}{3} + 2$

Part B: Absolute Value Inequalities

Problem 4

Solve: $|x + 2| < 6$

(a) Graph the solution on a number line.

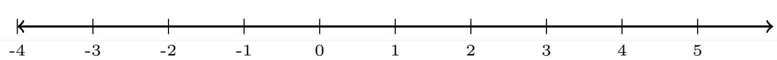


(b) Write the solution in interval notation.

Problem 5

Solve: $|3x - 1| \geq 8$

(a) Graph the solution on a number line.



(b) Write the solution in interval notation.

Problem 6

A machine part must be 50 mm long.

It is acceptable if the length differs by at most 0.8 mm.

Let L = measured length (in mm).

Write and solve an inequality for L.

State the acceptable range.

(a) Graph the solution on a number line.

(b) Write the solution in interval notation.

Part C: Parallel & Perpendicular Lines

Problem 7

Find the equation of the line through $(-3, 2)$ and parallel to $y = -3x + 5$

Problem 8

Find the equation of the line through $(4, 1)$ and perpendicular to $3x - y = 9$

Problem 9

Determine if the lines are parallel, perpendicular, or neither:

- Line 1: $y = \frac{2}{5}x + 3$
- Line 2: $y = -\frac{5}{2}x - 1$

Part D: Systems of Linear Equations

Problem 10

Solve the system using substitution:

$$y = 2x - 5$$
$$y = -x + 7$$

Problem 11

Solve the system using elimination:

$$\begin{aligned} 4x + 3y &= 10 \\ -4x + 5y &= 6 \end{aligned}$$

Problem 12

Classify the system (one solution / no solutions / infinitely many solutions):

$$\begin{aligned} y &= \frac{1}{2}x + 4 \\ 2y &= x + 8 \end{aligned}$$

Part E: Systems Applications

Problem 13: Motion Problem

Two trains leave the same station at the same time.

They travel in opposite directions.

Train A travels at 60 mph.

Train B travels at 80 mph.

After how many hours will they be 420 miles apart?

Problem 14: Word Problem

A movie theater sold tickets for adults and children.

There were 120 tickets sold in total.

Each adult ticket cost $15.

Each child ticket cost $8.

The total revenue was $1,470.

How many of each type of ticket was sold?